Total No. of Questions : 4]

SEAT No. :

PE-542

[Total No. of Pages : 2

[6578]-15

S.E. (E&TC/Electronics/Electronic Eng. (VLSIDesign & Technology)/(Electronics & Communication-Advanced Communication Technology) (Insem.)

DATA STRUCTURES

(2019 Pattern) (Semester - III) (204184)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
2) *Figures to the right indicate full marks.*
3) *Neat diagrams must be drawn wherever necessary.*
4) *Assume suitable data if necessary.*

Q1) a) Define Pointer, Explain pointer declaration, initialization, pointer arithmetic with suitable example? **[5]**

b) Explain call by value and call by Address for the example of swapping two numbers. **[6]**

c) Explain the difference between structure and union with suitable Example? **[4]**

OR

Q2) a) What is String? How to declare string in C? Write user defined function to calculate length of string? **[5]**

b) List different types of operators in C? Explain Bitwise operator in detail? **[5]**

c) Explain need of File Handling in C? Explain the difference between text and binary file? What are different file handling modes? **[5]**

Q3) a) Write step by step procedure for performing Binary search on following array [10, 22, 35, 40, 45, 50, 80, 82, 85, 90, 100] to search for the element 45. **[5]**

b) Explain Time and Space Complexity? Explain the significance of Big O, Big Theta, and Big Omega notations? **[5]**

c) Compare Bubble, Insertion, Selection, Merge, and Quick Sort with respect to stability, worst case time complexity, Adaptivity? **[5]**

P.T.O.

OR

Q4) a) Show All steps for performing insertion sort on [12, 15, 17, 11,9, 13, 18, 16]. **[5]**

b) What will be big(O) for the following code? Write main function to perform addition of *n* elements entered by user using array for the following code?

```
int sum (int arr [ ], int n)

{

  int i, total = 0;

  for (i = 0; i < n; i++) {

   total + = arr [i];

}

return total;

}
```

[5]

c) Compare Linear and Binary Search. Write an Algorithm to search the element in a list using Linear Search. **[5]**

❖❖❖